Huawei Access Node ELEC-ACC-57 Optical Transport Failure SOP (ALARM_7390)

Node: ELEC-ACC-57 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_7390

Description

The Huawei Access Node ELEC-ACC-57 at Electronic City Hub is experiencing an Optical Transport Failure, specifically an SFP/XFP degradation, fiber cut, or Rx/Tx power loss. This failure domain occurs due to the inherent fragility of the Optical Interface Module (OIM) and the susceptibility to environmental stressors such as temperature fluctuations, humidity, and vibration. The OIM, responsible for transmitting and receiving optical signals, is a critical component of the Huawei Access Node's Optical Transport Chain. However, its reliance on delicate optical fibers and sensitive laser diodes makes it prone to degradation and failure. Furthermore, the Huawei Access Node's Optical Transport Chain is designed to operate within specific power and temperature ranges, and deviations from these parameters can lead to OIM degradation and failure. The severity of this failure is classified as minor, but prompt attention is still required to minimize service disruptions and prevent further degradation.

Pre-requisite Checks

- Verify the site location and confirm that the failure is occurring at Electronic City Hub.
- Check the redundant link status to ensure that it is operational and can take over in case of a failure.
- Dispatch remote hands to the site to assist with physical checks and repairs.
- Gather information on recent environmental conditions, including temperature and humidity levels.

Potential Root Causes

Cause	Probability
SFP/XFP degradation due to environmental stressors	High
Fiber cuts or damage	Medium
Rx/Tx power loss due to OIM degradation	Low

Diagnostic Commands & Expected Outputs

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Command: display optical interface, Expected Output: Optical Interface Module information, including temperature, humidity, and power levels, Description: This command displays critical OIM information, allowing for the identification of potential issues and degradation. It is essential to check for any anomalies in temperature, humidity, and power levels, which may indicate OIM degradation or failure.

Command: display optical interface module, Expected Output: Detailed OIM information, including Rx and Tx power levels, Description: This command provides a more detailed view of the OIM, including Rx and Tx power levels. It is crucial to check for any abnormalities in these
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levels, which may indicate OIM degradation or failure.

Command: display alarm information, Expected Output: Alarm information, including ALARM_7390, Description: This command displays alarm information, including the specific ALARM_7390 code. It is essential to verify that the alarm is indeed related to the Optical Transport Failure and not another issue.
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Resolution Steps

Step 1: Step: Replace the Optical Interface Module (OIM), Command: replace optical interface module OIM-SLOT-1, Description: Replace the OIM with a new one to resolve the issue. Ensure that the new OIM is properly seated and secured before proceeding.

Step 2: Step: Check and replace the optical fibers, Command: replace optical fiber OIM-SLOT-1-FIBER-1, Description: Inspect the optical fibers for any damage or degradation. If damaged, replace the fibers with new ones to resolve the issue.

Step 3: Step: Verify the environmental conditions, Command: check environmental conditions, Description: Verify that the environmental conditions, including temperature and humidity levels, are within the recommended ranges. If not, take corrective action to ensure optimal operating conditions.

Post-Check Verification

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Command: display optical interface, Expected Output: Optical Interface Module information, indicating normal operation, Description: Verify that the OIM is operating within normal parameters, including temperature, humidity, and power levels.

Command: display optical interface module, Expected Output: Detailed OIM information, indicating normal operation, Description: Verify that the OIM is operating within normal parameters, including Rx and Tx power levels.

Command: display alarm information, Expected Output: No alarm information, including ALARM_7390, Description: Verify that the alarm is cleared and there are no further issues with the Optical Transport Failure.
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Roll-back Procedures

- Step: Restore the original OIM, Command: restore optical interface module OIM-SLOT-1, Description: Restore the original OIM to its previous state to resolve the issue.

- Step: Revert the optical fibers, Command: revert optical fiber OIM-SLOT-1-FIBER-1, Description: Revert the optical fibers to their previous state to resolve the issue.

- Step: Re-check the environmental conditions, Command: re-check environmental conditions, Description: Re-check the environmental conditions to ensure that they are within the recommended ranges.